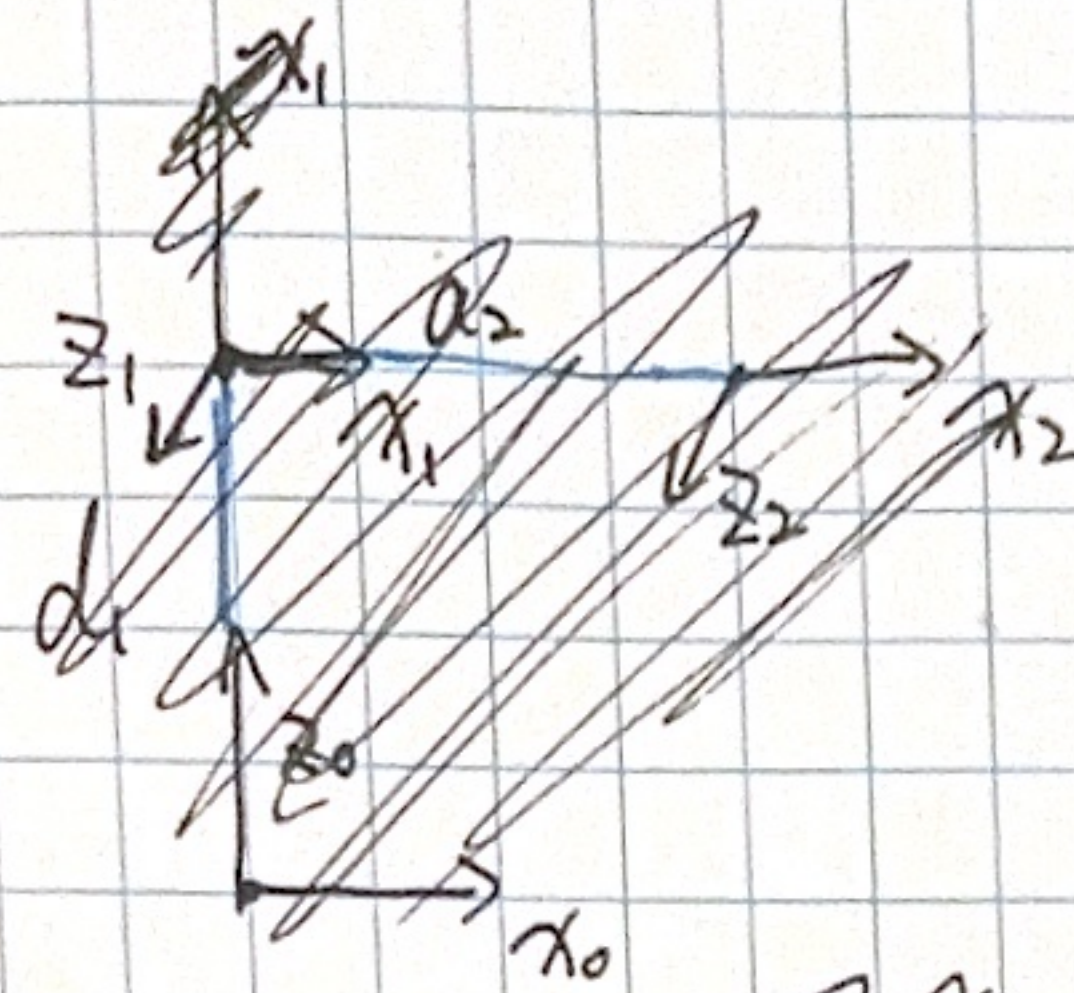
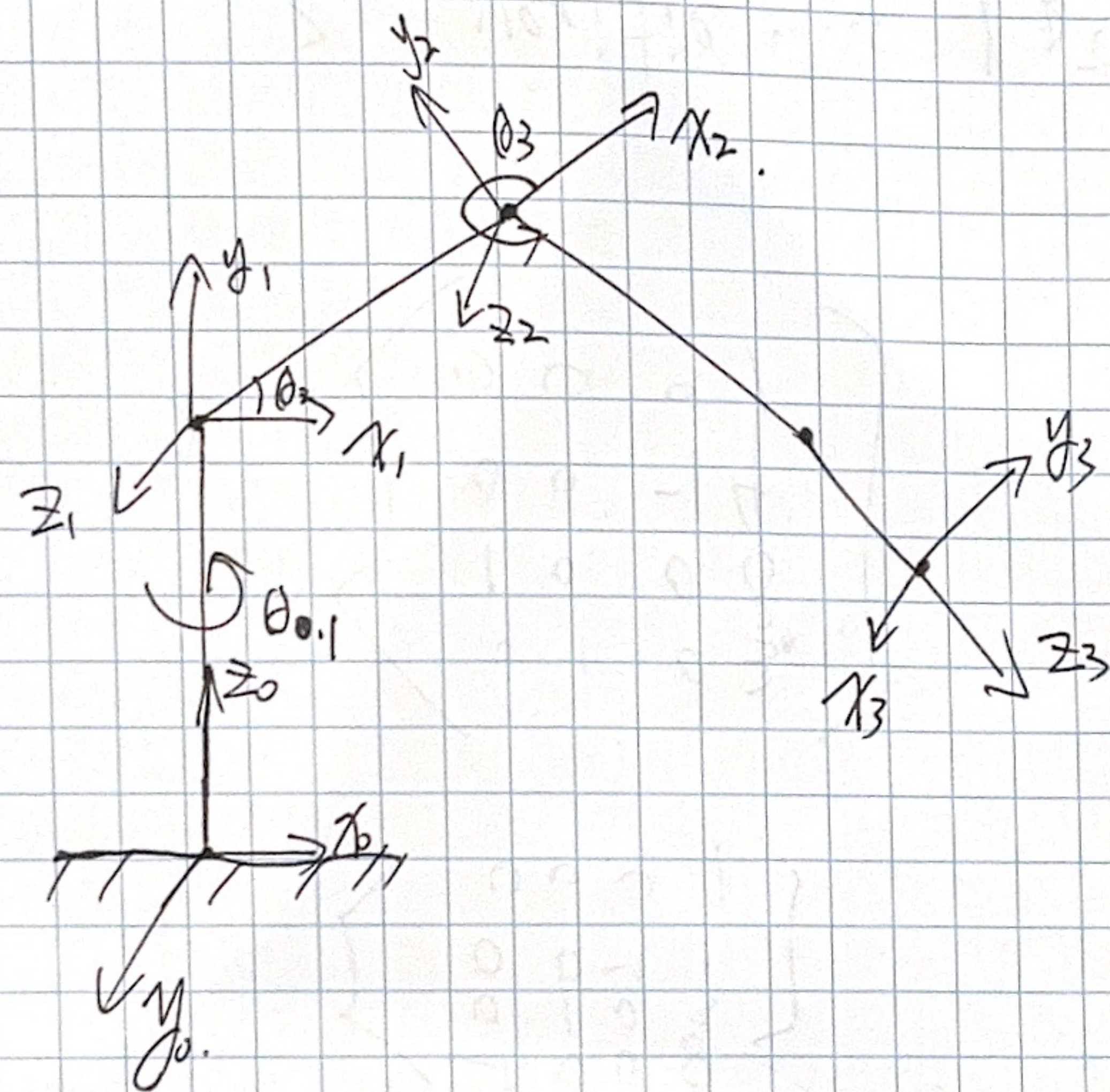
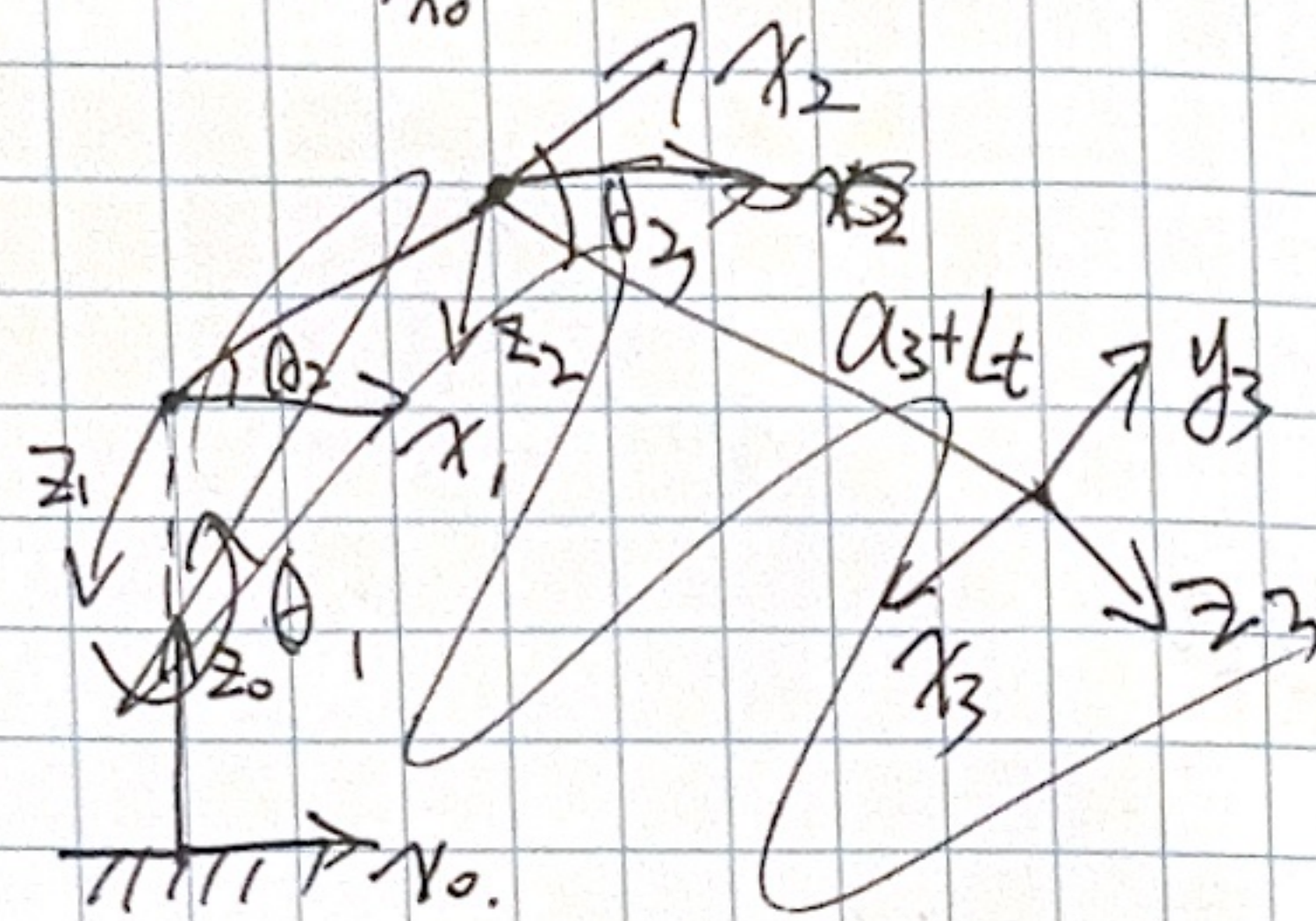


Q.



	θ_k	d_k	a_k	α_k
0	θ_0	d_1	0	$\pi/2$
1	θ_1	0	a_2	0
2	θ_2	0	$a_3 + L_t$	0

2. θ_3 joint rot



Q₂.

$$0 \text{ } I_{pick} = \begin{pmatrix} 0 & 0 & 0 & 8 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$1. \begin{pmatrix} -1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$2. \begin{pmatrix} 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 8 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

3. Verifie $R^T R = I$ $\det R = +1$

rotation is. $\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = R_z(-90^\circ)$